



PROBLEM SPACE

SOCIAL ROBOT CO-DESIGN CANVASES

What problem are you solving?

USER

Group(s) Who are the users? Are there supporting users? For example: students and teachers.	Characteristics What are the users like?	Needs What do the users need?
primary users Patients	Patient who has no medical knowledge and are often scared.	Patients need to know what to do that they are being taken care of.
secondary users ED Clinicians	Hard-working professionals who are often tired and stressed.	Clinicians need relevant medical information quickly to make relevant decisions.

Goal(s) What do the primary and secondary users want to accomplish?	
primary users Get help from a medical professional.	Go home after making a full recovery
secondary users Care for patients and adequately triage them. Make it through shift.	Maximize patient survival rate and minimize time between arrival and disposition.
SHORT-TERM	LONG-TERM

ROBOT NB: THIS IS FOR MY HCI

Task(s) What task does the robot perform?	
-Collects reason for visit, chief complaints and vitals from patient. - Allows patient to quickly self-triage. - Detect and alert nurses to critical patients without data entry.	- Increase the rate of correct triaging. - Uses ethically gathered patient information and usage metrics to minimize time to disposition.
SHORT-TERM	LONG-TERM

Advantages What advantages does using a robot bring (compared to a computer or human)?			
Social skills	User's emotional response	Personalization	Precise tasks
Data collection with sensors	Mobility	Environment manipulation	Connection to systems





ETHICAL CONSIDERATIONS

SOCIAL ROBOT CO-DESIGN CANVASES

Consider potential ethical problems, and potential solutions – both from the user's and robot's perspectives.
Consider the boxes to be guidelines: you don't need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

PROBLEM	USER	SOLUTION
	ROBOT	
User may damage screen.	Screen reinforced via Tempered Glass Shield	

Data security

Is the robot in a unique data collection position? How is the user's data protected?

PROBLEM	USER	SOLUTION
	ROBOT	
Collects all of the user's medical data.	Data is securely stored in a server on site and never leaves hospital grounds.	

Transparency

How does the robot share an accurate perception of its abilities, intentions and constraints, so the user can evaluate their trust in it?

PROBLEM	USER	SOLUTION
	ROBOT	
System may be unclear on why its asking for patient data leading to distrust.	It should be made abundantly clear in the UX on what the purpose of the data is.	

Equality across users

Robots' algorithms can be biased. A robot's appearance could reinforce harmful stereotypes. What are potential issues?

PROBLEM	USER	SOLUTION
	ROBOT	
Triage algorithms may systematically underperform on certain groups.	Triage performance different classes as well as the population must be tested	

Emotional consideration

People have been shown to form emotional attachments to robots, as if they were alive. Is this a potential problem?

PROBLEM	USER	SOLUTION
	ROBOT	

Behaviour enforcement

People could transfer their inappropriate behaviour, such as rudeness, from robots to humans. How is this mitigated?

PROBLEM	USER	SOLUTION
	ROBOT	



DESIGN GUIDELINES

SOCIAL ROBOT CO-DESIGN CANVASES

What things are important to consider in the robot's design?

Advantage guidelines

What advantages can the robotic solution have?
Think back to what you defined in the solution space canvas.

Ethical guidelines

What ethical considerations does the robot have?
Think back to what you defined in the ethics canvas.

ROBOT DIMENSIONS

Environment guidelines

What should the robot's context be like?
For example:

- If users are especially vulnerable, should it optimize for support?
- If the robot is part of a strict process, should it optimize for efficiency and security?

Form guidelines

What guides the design of the robot's outward qualities?
For example:

- Should the robot be designed to appear especially approachable, or more industrial?
- Should it be simple, or detailed?

Interaction guidelines

What guides the design of interaction?
For example:

- Is the interaction multimodal, or is one modality optimized for efficiency?
- Should the user feel empowered and lead the interaction, or does the robot provide safety via leadership?
- Is the goal of the interaction to complete a task, or explore?

Behaviour guidelines

What guides the design of the robot's behaviour?
For example:

- Should behaviour be simple, or sensitive to context?
- Does the robot have internal drivers, or does it react to external stimuli?
- Does the robot have social skills?





ROBOT DESIGN MVP

SOCIAL ROBOT CO-DESIGN CANVASES

It's time to design your robot MVP (Minimum Viable Product)! Remember the guidelines you defined.

Where and when

What place?
What time of day?
Does the place or time change?

Robot's role

Is the robot a friend? Teacher? Helper?
Something else?

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

Personality

Does the robot have specific characteristics?
Does it have emotional states, or needs?

Context-based behaviour

What external and environmental factors affect behaviour?
What data is used to adapt to context?

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

☐ voice	☐ touch	☐ movement	☐ screens
☐ sounds	☐ smell	☐ lights	☐ other _____
☐ gestures	☐ facial expressions	_____	_____

OUTPUT

☐ voice	☐ touch	☐ movement	☐ screens
☐ sounds	☐ smell	☐ lights	☐ other _____
☐ gestures	☐ facial expressions	_____	_____

Interaction flow

Describe the most important interaction of the robot.

Note: only fill the bottom row if your robot is teleoperated.

	BEFORE	DURING	AFTER
USER			
ROBOT			
ROBOT OPERATOR (optional)			





ENVIRONMENT

SOCIAL ROBOT CO-DESIGN CANVASES

What is the robot's context of operation?
You can use the "Ecosystem" canvas to dive deeper into this topic.

Where

What place?
Does it change?

At the ED entrance.

User(s)

Who is using the robot?

Patients use it to self-triage.

When

What time of day?
Does it change?

It should be active all day but
recommendations
on where patients should go may change
depending on if the department is open.

Secondary user(s)

Are there secondary users?
E.g. teachers that help students use a robot.

Triage nurse are secondary users who can triage
patients if they aren't able to self-triage.

Data collection

Does the robot collect data from its environment?
How is it stored?

The system collects data via both manual data
entry by patients and wearables that take vitals.



TRADE-OFF:

More data collection requires more attention to data security.

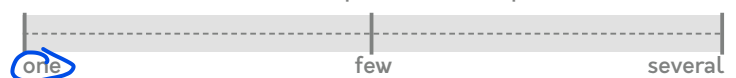
Simultaneous users

How many users should be able to use the robot
simultaneously?



TRADE-OFF:

More simultaneous users requires a more sophisticated robot.



External sensors and actuators

Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

The system has an attached wearable bracelet
that can measure various vital signs.

It also has speakers to ask questions via TTS.

Connection to systems

Is the robot connected to external systems, such as software,
databases, or other robots?
How does it use these systems?

The system is connected to the EHR and feeds
data into it.





FORM

SOCIAL ROBOT CO-DESIGN CANVASES

What are the robot's outward qualities? If an existing robot is used, are its qualities modified?

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

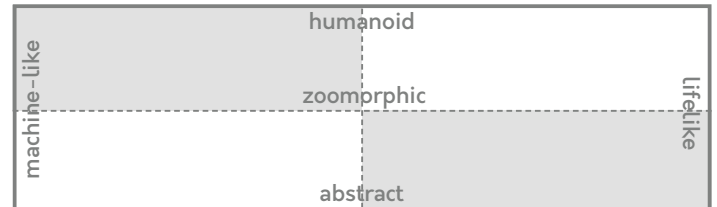
Appearance

Is the robot more machine or lifelike? Is it human-shaped, animal-shaped, or abstract?



TRADE-OFF:

Robots that appear more human and lifelike are expected to be more sophisticated in features.



Size

How big is the robot?



Character of movement

What is the robot's movement like?



Voice & sounds

Does the voice have a gender or an age? What are pitch, speed and prosody like? Is the voice always the same?
Does the robot make sounds: music, "beep"s, animal noises?
When are these sounds heard?

Mobility

Does the robot move across space? Does it move in place?

Visual cues

Does the robot have expressions, lights, a screen or other visual elements?

Touch & smell sensations

Is the robot soft or rough, warm or cold?
How does the robot smell?
Touch and smell are especially important in close interactions.





INTERACTION

SOCIAL ROBOT CO-DESIGN CANVASES

How does the robot interact with the user?
You can use the “Experience Flow” canvas to dive deeper into this topic.

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

☐ movement	☐ screens
☐ voice	☐ touch
☐ sounds	☐ lights
☐ gestures	☐ smell
☐ facial expressions	☐ other _____

OUTPUT

☐ movement	☐ screens
☐ voice	☐ touch
☐ sounds	☐ lights
☐ gestures	☐ smell
☐ facial expressions	☐ other _____

Interaction flow

Describe the most important interaction of the robot.
Note: only fill the bottom row if your robot is teleoperated.

	BEFORE	DURING	AFTER
USER			
ROBOT			
ROBOT OPERATOR (optional)			

Situation flow

How defined is the situation where the interaction takes place?
Does the user always enter and exit at the same point?

predefined	flexible	freestyle
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Leadership

Who initiates the interaction? Who determines what happens next?

robot-led	mutual / alternate	user-led
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Goal

What is the user's goal in the interaction? What describes the interaction?

task completion / informative	both	explorative
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Robot's name

Does the robot have a name which is used during interaction?



TRADE-OFF:

A robot with a name, creates more emotional bond.





BEHAVIOUR

SOCIAL ROBOT CO-DESIGN CANVASES

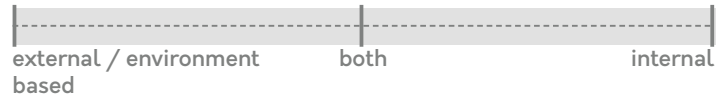
What factors guide the robot's behaviour?

Robot's role

Is the robot a friend? Teacher? Helper?
Something else?

Motivation

How is the robot's behaviour motivated? Is it based on external data, internal models such as personality, or both?



Mode of operation

Is the robot operating by itself, or is a human affecting behaviour? Is a human in full control?



TRADE-OFF:

A human-operated robot requires a good user interface, an autonomous robot requires a good control logic.



Social skills

How good are the robot's social skills: does it greet a new person and ask their name? Does it follow people with its gaze?



TRADE-OFF:

Extensive social skills require a more sophisticated robot.



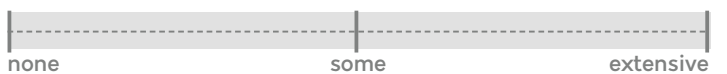
Contextual adaptation

Does the robot's behaviour vary according to context, e.g. by weather or time of day?



TRADE-OFF:

More contextual adaptation requires a more sophisticated robot.



Context-based behaviour

What external and environmental factors affect behaviour?
What data is used to adapt to context?

Personalization

Does the robot behave differently toward different people?
Does it need to remember people, and store their data?



TRADE-OFF:

More personalization requires more personal data from the user.

Personality

Does the robot have specific characteristics?
Does it have emotional states, or needs?



TRADE-OFF:

More personality creates more emotional bond.

Social behaviours

What social behaviours does the robot exhibit?



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THRI paper:

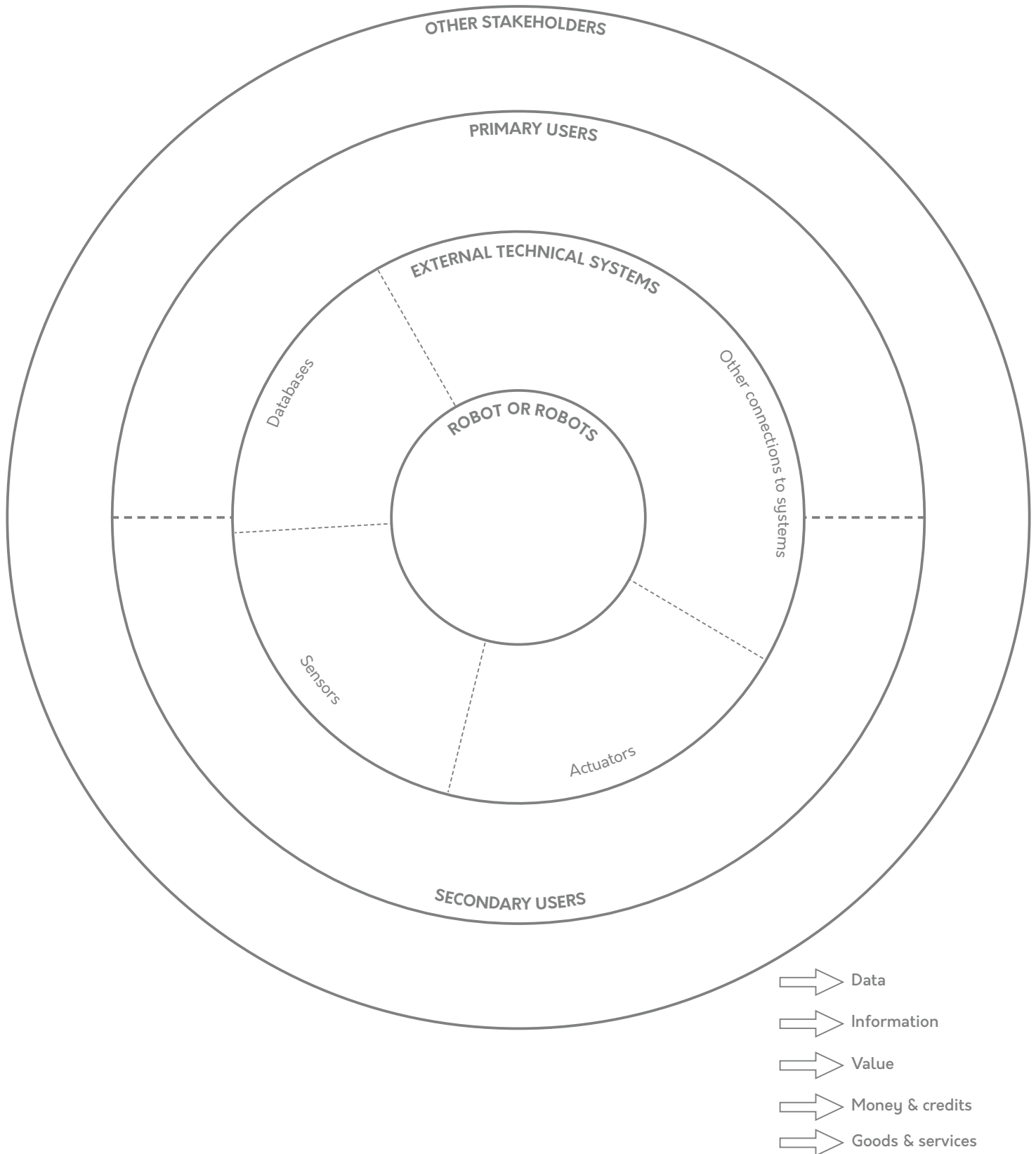


SERVICE ECOSYSTEM

SOCIAL ROBOT CO-DESIGN CANVASES

What stakeholders does the robot's operation involve?
Draw sectors for different stakeholders.

Select a color for each of the resources, and draw arrows to show their flow from stakeholder to stakeholder.





EXPERIENCE FLOW

SOCIAL ROBOT CO-DESIGN CANVASES

Describe the most important interaction of the robot.
Note: only fill the bottom row if your robot is teleoperated.

USER

FEELING e.g. confused		
THINKING e.g. "I need help."		
DOING e.g. pushes button		
DOING e.g. says "Hello!"		
SENSOR INPUT e.g. sees user's face		
CONNECTION TO SYSTEMS e.g. records data in database		
DOING e.g. controls robot's arm		
BEFORE	DURING	AFTER

ROBOT

ROBOT
OPERATOR
(optional)



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DESIGN PATH

SOCIAL ROBOT CO-DESIGN CANVASES

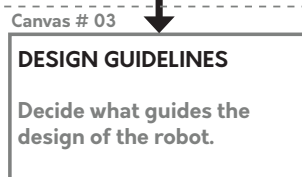
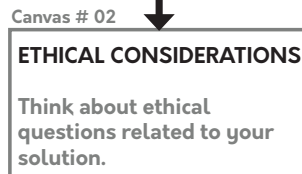
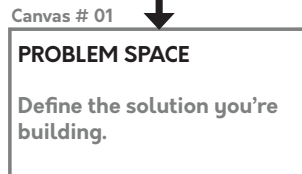
How to choose your canvases

Canvases:



START HERE

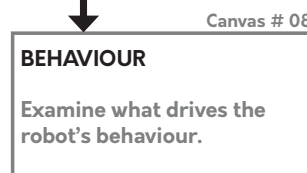
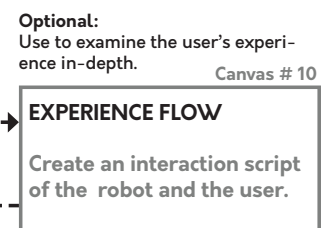
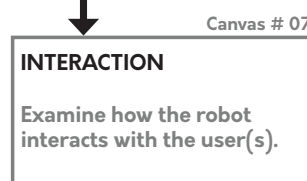
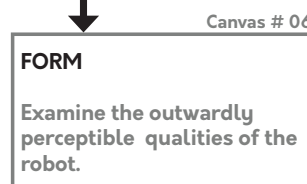
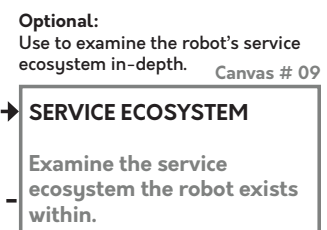
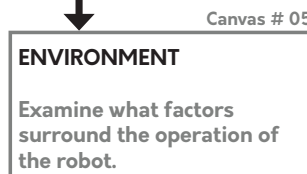
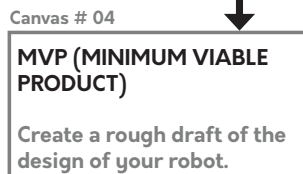
PHASE 1: PROBLEM SPACE



PHASE 2: DESIGN GUIDELINES

PATH 1:
A quick first draft of the robot design. Choose to create first ideas, or to choose between ideas.

PATH 2:
In-depth design of the robot and its four dimensions. Choose to create the final product design.



PHASE 3: SOLUTION SPACE

FINISHED



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